Algebra II with Trigonometry Chapter 7.3

Study guide

Saved tutoring response with MathJax rendering. Original PDF file: 4592201_alg-2trig-h-chp-7-3-note-docx.pdf

Certainly! Below is a **self-contained study guide** based on the provided document:

Chp 7.3: Double-Angle, Half-Angle, and Product-Sum Formulas

1. Short Summary

This chapter covers key trigonometric formulas used to simplify and evaluate trigonometric expressions involving double angles, half angles, and products of sines and cosines:

- **Double-Angle Formulas**: Find the value of a trig function at $2x$ from its value at x.
- **Half-Angle Formulas**: Find the value of a trig function at $\frac{x}{2}$ from its value at x.
- **Product-Sum and Sum-Product Formulas**: Rewrite products of sines/cosines as sums or vice versa, useful for simplifying complicated expressions.

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2. Core Definitions, Theorems, and Formulas

I. Double-Angle Formulas

\[\begin{align} \sin(2x) &= 2\sin x \cos x \ \cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 \ \tan(2x) &= \frac{2\tan x}{1 - \tan^2 x}

\end{align} \]

II. Half-Angle Formulas

\[\begin{align} \sin \frac{u}{2} &= \pm \sqrt{\frac{1 - \cos u}{2}} \ \cos \frac{u}{2} &= \pm \sqrt{\frac{1 + \cos u}{2}} \ \tan \frac{u}{2} &= \frac{1 - \cos u}{\sin u} = \frac{\sin u}{1 + \cos u} \end{align} \] (Sign depends on the quadrant.)

Formulas for Lowering Powers

\[\begin{align} \sin^2 x &= \frac{1 - \cos 2x}{2} \ \cos^2 x &= \frac{1 + \cos 2x}{2} \ \tan^2 x &= \frac{1 - \cos 2x}{1 + \cos 2x} \end{align} \]

III. Product-Sum and Sum-Product Formulas

Product-to-Sum

\[\begin{align} \sin u \cos v &= \frac{1}{2}\left[\sin(u + v) + \sin(u - v)\right] \ \cos u \sin v &= \frac{1}{2}\left[\sin(u + v) - \sin(u - v)\right] \ \cos u \cos v &= \frac{1}{2}\left[\cos(u + v) + \cos(u - v)\right] \ \sin u \sin v &= \frac{1}{2}\left[\cos(u - v) - \cos(u + v)\right] \end{align} \]

Sum-to-Product

\[\begin{align} \sin x + \sin y &= 2 \sin\left(\frac{x + y}{2}\right)\cos\left(\frac{x - y}{2}\right) \ \sin x - \sin y &= 2 \cos\left(\frac{x + y}{2}\right)\sin\left(\frac{x - y}{2}\right) \ \cos x + \cos y &= 2 \cos\left(\frac{x + y}{2}\right)\cos\left(\frac{x - y}{2}\right) \ \cos x - \cos y &= -2 \sin\left(\frac{x + y}{2}\right)\sin\left(\frac{x - y}{2}\right) \end{align} \]

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3. Worked Study Guide: Step-by-Step Explanation

Step 1: Recognize Which Formula to Use

- **Double-Angle**: When you have $\sin(2x), \cos(2x), \tan(2x)$.

- **Half-Angle**: When you have $\sin\left(\frac{x}{2}\right), \cos\left(\frac{x}{2}\right), \tan\left(\frac{x}{2}\right)$.
- **Product-Sum/Sum-Product**: When dealing with multiplication or addition/subtraction of trig functions with different arguments.

Step 2: Double-Angle Example

Find $\sin(2x)$ given $\sin x = \frac{3}{5}$, x in Quadrant II.

- $\sin(2x) = 2\sin x \cos x$
- First, find $\cos x$: In Quadrant II, $\cos x < 0$.
- $\sin x = \frac{3}{5}$ so $\cos x = -\sqrt{1 - (\frac{3}{5})^2} = -\frac{4}{5}$
- $\sin(2x) = 2 \cdot \frac{3}{5} \cdot \left(-\frac{4}{5}\right) = -\frac{24}{25}$

Step 3: Half-Angle Example

Find $\cos\left(\frac{x}{2}\right)$ if $\cos x = -\frac{4}{5}$ and x is in quadrant III (so $\frac{x}{2}$ is in quadrant II, cos is negative).

$$\cos\left(\frac{x}{2}\right) = -\sqrt{\frac{1 + (-\frac{4}{5})}{2}} = -\sqrt{\frac{1 - \frac{4}{5}}{2}} = -\sqrt{\frac{1/5}{2}} = -\sqrt{\frac{1}{10}} = -\frac{1}{\sqrt{10}}$$

Step 4: Product-to-Sum Example

Simplify $\sin 2x \sin 3x$:

$$\sin 2x \sin 3x = \frac{1}{2}[\cos(2x - 3x) - \cos(2x + 3x)] = \frac{1}{2}[\cos(-x) - \cos(5x)]$$

Step 5: Lowering Powers Example

Rewrite $\sin^4 x$ in terms of the first power of cosine:

- First, $\sin^2 x = \frac{1 - \cos 2x}{2}$
- Thus, $\sin^4 x = (\sin^2 x)^2 = \left(\frac{1 - \cos 2x}{2}\right)^2 = \frac{1 - 2\cos 2x + \cos^2 2x}{4}$
- Now, lower power of $\cos^2 2x$: $\cos^2 2x = \frac{1 + \cos 4x}{2}$, substitute in and simplify.

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4. Practice Problems with Answers

1. Simplify $\sin(2x)$ if $\sin x = \frac{1}{2}$, x in Quadrant I.

> **Answer:** $\cos x = \sqrt{1-(1/2)^2} = \sqrt{3}/2$
> $\sin(2x) = 2 \cdot (1/2) \cdot (\sqrt{3}/2) = \sqrt{3}/2$

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2. Express $\cos^2 x$ in terms of $\cos 2x$.

> **Answer:** $\cos^2 x = \frac{1+\cos 2x}{2}$

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3. Simplify $\sin x + \sin 3x$ using a sum-to-product formula.

> **Answer:**

$$\sin x + \sin 3x = 2\sin\left(\frac{x+3x}{2}\right)\cos\left(\frac{x-3x}{2}\right) = 2\sin(2x)\cos(-x) = 2\sin($$

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4. Write $\sin x \cos 2x$ as a sum.

> **Answer:**

$$\sin x \cos 2x = \frac{1}{2}[\sin(x+2x) + \sin(x-2x)] = \frac{1}{2}[\sin(3x) + \sin(-x)] = \frac{1}{2}(\mathrm{si}$$

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5. Use the half-angle formula to find $\tan\left(\frac{x}{2}\right)$ if $\sin x = \frac{4}{5}$ and x is in Quadrant II.

> **Answer:** $\cos x = -\sqrt{1-(4/5)^2} = -3/5$
> $\tan\left(\frac{x}{2}\right) = \frac{1-\cos x}{\sin x} = \frac{1-(-3/5)}{4/5} = \frac{1+3/5}{4/5} = \frac{8/5}{4/5} = 2$

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5. Assumptions or Ambiguities

- For half-angle, **signs** depend on the quadrant—always check where the angle $\frac{x}{2}$ lies.
- Some expressions like $\sin^4 x$ require repeated use of the lowering power formula.

- Problems ask for “exact value,” so answers should be in simplest radical form or rationalized.
- If a variable is unspecified, assume general values or positive roots unless a quadrant is stated.
- Quadrant information is crucial for determining sign in half/double angle values.

End of study guide. If you need worked solutions for any of the specific example problems from the document, let me know!